

EDUCATION

Stony Brook University

Stony Brook, NY

Expected May 2030

- B.S. Computer Science, Sophomore Standing
- Honors College, Honors in Computer Science Program

Stuyvesant High School

New York, NY

June 2026

GPA: 97.64/100

- Excellence in Mathematics Award
- Relevant Coursework: Linear Algebra, AP Calculus BC, Cybersecurity, AP Physics C

TECHNICAL SKILLS

Languages: C, SQL, Python, MATLAB, C#, Java

Tools: Git, MongoDB, Supabase / PostgreSQL, KiCAD, LaTeX

Frameworks: Next.js, Node.js, HTML/CSS

AWARDS & HONORS

- AIME Qualifier (3x)
- CEMC Euclid/Fermat School Winner (2025)
- Gold President's Volunteer Service Award

PROFESSIONAL EXPERIENCE

Memorial Sloan Kettering Cancer Center

New York, NY

Student Intern, Medical Physics

May 2025 - November 2025

- Fine-tuned and validated a CNN-based deformable image registration (DIR) model for aligning cone-beam CT scans to planning CTs for lung cancer treatment monitoring, achieving validation DICE score over 0.75
- Experimented with a multi-head attention architecture for DIR, reaching parity with CNN and VoxelMorph baselines
- Wrote MATLAB and Python code to preprocess and validate training datasets
- Ran training and inference on HPC clusters and configured CUDA environments for training compatibility

Summer Intern, Medical Physics

July 2024 - August 2024

- Wrote C# and Python preprocessing code for head/neck cancer CT segmentation training for treatment tracking
- Leveraged HPC clusters and IBM LSF (bsub) for job scheduling

Hack Club

Shelburne, VT and Remote

Hardware Engineering Intern

May 2022 - March 2024

- Designed and prototyped production PCBs for Blot (2D CNC drawing machine) and Sprig (RP2040-based handheld gaming console), including motor control and USB-C PD boards; Sprig reached a 600+ unit production run
- Published open-source reference designs adopted by the maker community (NXP/NTAG I2C NFC board, USB hub with USB-C implementation guide) and mentored dozens of high schoolers through first-time PCB design

Logistics and Operations Intern

July 2021 - August 2023

- Built a mailing address database using USPS Intelligent Mail Barcode (IMb), enabling no-cost trackable postage
- Reduced domestic shipping costs up to 60% and international costs up to 80%, including devising alternate routing through regional trade disruptions
- Integrated the Shippo API into a Slack-based shipping agent to automate label generation and cost comparison
- Planned and managed on-the-ground logistics with organizers and sourced affordable and high-quality merchandise for in-person hackathons in San Francisco (approx. 150 attendees), Vermont, and New York

PROJECTS

Personal Site and Photography Portfolio | Next.js, Tailwind CSS, Cloudinary CDN | GitHub 2026

- Designed a performant Next.js photography site with Cloudinary CDN delivery and EXIF-based filtering, cutting image load times and simplifying browsing across collections
- Built a secure authenticated admin dashboard (NextAuth, Bcrypt) backed by Supabase PostgreSQL, enabling non-technical content updates without redeploying the site

Component Inventory Manager | Next.js, PostgreSQL (Supabase), NextAuth | GitHub 2026

- Implemented a full-stack Next.js/Supabase system to manage electronic components, MPNs, and distributor data, with a barcode-scanner-compatible search tool for fast lookups
- Generated thermal-printable asset ID labels to tag and organize components, streamlining inventory retrieval